Q1. General recipe for solving $V(X^I) = X^M$ for X^I

Write the problem as a root-finding problem:

$$F(X^I) \equiv V(X^I) - X^M = 0.$$

Solve it with Newton–Raphson:

1. Choose an initial guess X_0^I.
2. Compute the Jacobian $J(X_k^I)$ with $J_{ij} = \frac{\partial V_i}{\partial X_j^I}$, either analytically or by bumping.
3. Solve the linear system $J(X_k^I)\,\Delta = -F(X_k^I)$. Do not form J^{-1} explicitly.
4. Update $X_{k+1}^I = X_k^I + \Delta$.
5. Stop when $\| F(X_k^I) \| < \varepsilon$, or when $\| \Delta \|$ is below a tolerance.

Practical points:

- Use a sensible initial guess, such as the previous day's solution or a flat model.
- Add bounds, damping or a line search so the iteration stays in the valid parameter domain.
- If J is close to singular, use Levenberg–Marquardt, which solves $(J^T J + \lambda I)\Delta = -J^T F$.
- If there were more market quantities than parameters, the problem would become a least-squares fit, $\min_{X^I} \| F \|^2$. With equal dimensions we solve exactly.
- The solution is locally unique when J is invertible at the root.

Q2. Calculating $\frac{dX^I}{dX^M}$ at the optimal X^I

Define the calibration residual

$$F(X^I, X^M) = V(X^I) - X^M = 0.$$

At the solution, X^I is an implicit function of X^M, so $V\big(X^I(X^M)\big) = X^M$ for all X^M near the market. Take the total differential of $F = 0$:

$$\frac{\partial F}{\partial X^I}\, dX^I + \frac{\partial F}{\partial X^M}\, dX^M = 0 \quad \Rightarrow \quad \frac{dX^I}{dX^M} = -\left(\frac{\partial F}{\partial X^I}\right)^{-1} \frac{\partial F}{\partial X^M}.$$

Here $\partial F / \partial X^I = \partial V / \partial X^I = J$ and $\partial F / \partial X^M = -I$. The minus signs cancel, so there is no sign flip in the final answer:

$$\frac{dX^I}{dX^M} = J^{-1}, \qquad J = \frac{\partial V}{\partial X^I}\bigg|_{X^I = X^{I*}}.$$

This is the implicit function theorem. J is the same Jacobian Newton already computed in the last iteration.

How J is built when V goes through the model. Because V acts on the parametric model $M(X^I)$, we have $V = V\big(M(X^I)\big)$, and the Jacobian factorises by the chain rule:

$$J = \frac{\partial V}{\partial M} \cdot \frac{\partial M}{\partial X^I}.$$

In practice, compute $\partial M / \partial X^I$ analytically (or with AAD), or by bumping each X_j^I.

Conditions for the result to hold.

- J is invertible at X^{I*}. If J is singular, the result fails.
- V is continuously differentiable near the solution.
- The result is local: it is valid for small moves in X^M, and it assumes an interior optimum (no parameter bound is active).
- V does not depend on X^M. The calibration instruments are held fixed.
- J^{-1} is only as accurate as the solve. If Newton stopped at tolerance ε, then J is evaluated at an approximate root.

Sensitivity of any downstream quantity. For a scalar $f(X^I)$, for example a PV or an interpolated vol, the sensitivity to the market inputs is

$$\frac{df}{dX^M} = \left(\frac{\partial f}{\partial X^I}\right)^T J^{-1}.$$

Compute this by solving $J^T w = \nabla_{X^I} f$ once and reading off w. Then one solve gives the sensitivity to every market input, without forming J^{-1}.

Q3. Application: calibrating a vol interpolator to three OTM options

Market data. We observe undiscounted OTM prices P_1, P_2, P_3 at strikes $K_1 < K_{atm} < K_3$. Typically P_1 is a put, P_2 is the ATM option, and P_3 is a call. Let $\sigma(K;\theta)$ be the vol interpolator, with $\theta \in \mathbb{R}^3$.

Mapping to the general framework.

General	This problem
X^I	θ (interpolator parameters)
X^M	(P_1, P_2, P_3) (market prices)
$V_i(X^I)$	$\text{Black}(F, K_i, T, \sigma(K_i;\theta), \text{cp}_i)$ (model price at K_i)

How to find θ. Solve the three equations

$$V_i(\theta) = \text{Black}\big(F, K_i, T, \sigma(K_i;\theta)\big) = P_i, \qquad i = 1,2,3,$$

with Newton–Raphson as in Q1. The Jacobian factorises by the chain rule:

$$J_{ij} = \frac{\partial V_i}{\partial \theta_j} = \text{Vega}_i \cdot \frac{\partial \sigma(K_i;\theta)}{\partial \theta_j} \quad \Leftrightarrow \quad J = \text{diag}(\text{Vega}) \cdot \frac{\partial \sigma}{\partial \theta},$$

where $\text{Vega}_i = \partial\,\text{Black}/\,\partial\sigma$ at K_i.

An equivalent two-step route is to first invert each price to its market implied vol σ_i^{mkt} by a one-dimensional root solve, then solve $\sigma(K_i;\theta) = \sigma_i^{mkt}$ for θ. The two routes give the same θ.

Why the solution is unique.

- Vega is positive for an option with time value, so price and implied vol are one-to-one at each strike.
- Using OTM options is standard: they carry the time value, are the liquid quotes, and have well-behaved vega.
- With three parameters, three quotes, and a non-degenerate parametrisation, $\partial\sigma/\,\partial\theta$ is invertible. J is then invertible and the root is locally unique.

How $\sigma(K)$ changes with θ. The direct sensitivity at a fixed calibrated θ comes from the interpolator, analytically or by bump:

$$\frac{\partial \sigma(K;\theta)}{\partial \theta_j}.$$

How $\sigma(K)$ changes with the market prices. Combine Q2 with the chain rule:

$$\frac{d\theta}{dP} = J^{-1} = \left(\frac{\partial \sigma}{\partial \theta}\right)^{-1} \text{diag}\left(\frac{1}{\text{Vega}}\right), \qquad \frac{d\sigma(K)}{dP_i} = \frac{\partial \sigma(K;\theta)}{\partial \theta} \cdot \frac{d\theta}{dP_i}.$$

So $\Delta\sigma(K) \approx \frac{\partial \sigma(K;\theta)}{\partial \theta} J^{-1}\,\Delta P$. A shock to one quoted price moves the whole smile, not just the vol at that strike.

What θ might be, intuitively. With three quotes and three parameters, the natural choice is for θ to describe the smile in trader terms:

Parameter	Trader meaning	Effect on $\sigma(K)$	Driven mainly by
θ_1	ATM vol level	Parallel shift of the whole smile	P_2 (ATM)
θ_2	Skew / risk reversal	Tilts the smile: puts vs. calls, roughly $\sigma(K_3) - \sigma(K_1)$	$P_3 - P_1$
θ_3	Curvature / butterfly	Raises or lowers both wings relative to ATM, roughly $\frac{1}{2}\big(\sigma(K_1) + \sigma(K_3)\big) - \sigma(K_2)$	$P_1 + P_3 - 2P_2$

This matches the familiar parameters of SABR or SVI: level (α), skew (ρ) and smile curvature (ν). With this parametrisation $\partial\sigma/\,\partial\theta$ is close to diagonal, so J is well-conditioned and the sensitivities $d\theta/dP$ are easy for a trader to interpret. “ATM vol up one point” becomes a parallel shift. A change in risk reversal tilts the smile. A change in butterfly bends the wings.